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Enhancing social acceptance of women's mobility for equitable transport: Behavioural theories driven approach

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ABSTRACT

Women's mobility is hindered by socio-cultural norms, especially in developing countries, which limits their choices and opportunities for education and employment. This study aims to investigate social acceptance of women's mobility and promote equitable transport services for women in Pakistan. A survey questionnaire was developed, incorporating the concepts of the Theory of Planned Behaviour and Norm Activation Model, and was administrated online for data collection. A total of 478 responses were gathered and analysed utilizing the multivariate Structural Equation Modelling technique. The model results reveal that attitudes, subjective norms, and personal norms, are significant determinants of the social acceptance of women's mobility. While perceived behavioural control does not emerge as a significant predictor of the social acceptance, it is a significant positive predictor in influencing perceived transport policies, which play a mediating role. Road users' awareness and sense of responsibility are important in enhancing women's mobility by improving their security and reducing harassment on roads and public transport. Society's confidence and social support for independent women travellers, as well as acceptance of women as drivers, are vital in improving their access to schools and workplaces. Other supportive transport policies, including preferred parking, dedicated facilities, and specialized carpooling programs, are recommended. The outcomes of this study hold significant implications for transport planners and decision-makers, offering valuable insights for the provision of equitable transport services for women.

1. Introduction

1.1. Restrictions on women's mobility

Women's mobility is hindered by various socio-cultural norms, especially in developing countries. Women's domestic responsibilities and concerns about personal safety and security greatly influence their ability to travel independently. Women have limited options for affordable and reliable transport due to social, cultural, religious, and infrastructural constraints (Bernard et al., 2022). Conservative gender norms limit women's mobility and access to education and employment (Oktay et al., 2023). In some extreme cases, women are not even allowed to travel alone because of security, harassment, and social concerns (Zulfiqar, 2020), and they also face restrictions in obtaining a driving license to drive alone (Batool and Carsten, 2018). As a result, they often rely on their male family members to drive them around in private vehicles for their daily travel needs. These unjust travel restrictions unfairly limit women's mobility and hinder their access to basic needs.

Increasing the share of women's mobility is crucial for creating sustainable and liveable societies that provide equal opportunities for education, healthcare, and employment (Osmond and Woodcock, 2015). Many countries worldwide, both developed and developing, are actively working towards implementing gender-informed transport policies and increasing the share of women for equitable transport opportunities (Keyes, 2023). For example, Australia has set a target to double the utilization of public transport by women and improve the

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accessibility and land-use and transport integration by 2031 (Department of Transport, The Government of Western Australia, 2011). A study conducted in Tanzania emphasized the importance of enhancing women's mobility in rural areas by implementing a standardized pricing system and improving infrastructure, which should be included in the national transport policy (Ngusa, 2023). Equal opportunities for women to use public transport would enable them to make informed decisions about their lives (Kurshitashvili et al., 2021). Research indicates that the primary factors influencing women's travel preferences and utilization of public transport are; the quality of public transport services, priority treatment of women in services, permission to drive alone, and supportive and conductive road environments. According to Adeel et al. (2017), the share of women's travel is only 20 % in the travel market in Pakistan, significantly lower than that in many developed countries. Besides, many other studies have revealed that women's mobility can be affected by various factors, such as social, cultural, economic, and infrastructure contexts, which can vary across different regions (e.g., Mcllroy, 2023; Borker, 2022a; Collins and Tisdell, 2002). For instance, a recent study conducted in India reveals a significant difference between men's and women's mobility, with 53 % of women reporting that they do not travel outside of their homes (Goel, 2023). Other studies have revealed that women's mobility is significantly different, albeit suppressed, compared to their counterparts (Rosenbloom, 2004; Babinard and Scott, 2011). However, the exploration of women's mobility in developing countries remains largely unaddressed (Patnala et al., 2023), and it is anticipated that the share of women's mobility is significantly lower due to their reduced access to various mobility options and socio-cultural constraints (Gossen and Purvis, 2005).

Pakistan is one of the developing country that faces challenges related to women's mobility and transport equity. Women in Pakistan often travel infrequently due to concerns about privacy and safety in open spaces and public transport. Sometimes, their travel must be authorized by the household head, and in some cases, they require male family members to accompany them (Zulfiqar, 2020). Consequently, the marital status of women significantly impacts their mobility, with unmarried women facing the most restrictions in all forms of travel. They also encounter difficulties in accessing basic needs, such as education and healthcare facilities, even within their villages (Khan, 1999). Therefore, it is crucial to comprehend the factors influencing women's travel decisions and recognize the obstacles and difficulties they encounter. This understanding is essential for formulating related transport policies and interventions aimed at enhancing their accessibility (Borker, 2022b).

1.2. Factors influencing women's travel decisions

Several factors contribute to shaping women's travel preferences. Because of income differences, women are less likely to travel compared to men, as trip costs and lengths tend to increase with higher incomes (Mahadevia and Advani, 2016). Several studies have also revealed that women are more inclined to walk for short distances and less likely to use public transport compared to men. These preferences are influenced by socio-cultural norms and gender-specific barriers (Adorean et al., 2023; Pojani et al., 2017; Goel et al., 2022). With no doubt, women have different travel needs compared to men, influenced by safety concerns, socio-cultural constraints, and differential access to private transport. Furthermore, transport needs may also vary among different groups of women according to their age, and class/ethnicity (Loukaitou-Sideris, 2016). It is noted that specific socio-cultural aspects are influential in shaping the mobility and activity choices of women in developing countries. For instance, age and household income have a negative impact on women's mobility, and marital status notably influences their travel behaviour, with unmarried women being less likely to travel alone (Adeel et al., 2017). A study in Saudi Arabia indicates that socio-economic status, societal norms, and family opposition are the primary factors restricting women's mobility (Macias-Alonso et al., 2023). Younger and single women, with exposure to foreign cultures, and employment, significantly influence women's willingness to drive (Al-Garawi and Kamargianni, 2022). Unsafe infrastructure, driving environments, and vehicle design contribute to women's increased susceptibility to road accidents (Borker, 2022b). In addition, women's motivations and intentions to travel alone are influenced by their personal values and factors, including a sense of self-fulfilment and accomplishment, fun and enjoyment in life, excitement, freedom and flexibility, self-empowerment, and liberty (Hosseini et al., 2022; Teng et al., 2023). Gender-based stereotype behaviour, socio-cultural stigma, inadequate transport system settings, and a male-oriented transport sector hinder the easy access of women to public transport and harassment-free travel in urban spaces (Gardner et al., 2017; Noor and Iamtrakul, 2023). Research on public transport services in Mexico city illustrated that gender-based violence limits women's mobility and underpins inequality (Graglia, 2016). They encounter multiple challenges when using public transport because of gender inequality, sometimes their public transport experiences are a necessity instead of a choice, and these challenges are class-based and triggered by gender inequalities (Keysan et al., 2022). Unsafe and insecure transport alternatives hinder women's access to basic services (Zulfiqar, 2020). Socio-cultural contexts also exert a noteworthy influence on women's mobility, such as the social code of honour and unwanted interactions with men on public transport (Adeel et al., 2017), which are critical in shaping women's mobility (Chamseddine, Ait Boubkr, 2021). The direction of transport policies and the factors governing them are essential considerations when planning for women's mobility needs (Akyelken, 2020).

1.3. Theory of Planned Behaviour and Norm Activation Model

To examine transport policies and their socio-cultural aspects, this study adopts a combined approach of the Theory of Planned Behaviour (TPB) and the Norm Activation Model (NAM) to explore the social acceptance of women's mobility. The TPB, initiated by Ajzen as an extension of the Theory of Reasoned Action (Ajzen, 1985, 1991; Fishbein and Ajzen, 1975), has main components such as individual attitudes, Subjective Norms (SN), Perceived Behavioural Control (PBC), intentions and behaviours. The TPB is based on the proposition that an individual's behaviour is a direct function of intentions and PBC. Intentions are shaped by attitudes, SN, and PBC (Ajzen, 1991). Many studies have adopted the TPB for explaining travel behaviours (Ali et al., 2023; Bamberg et al., 2007; Chen and Chao, 2011). The NAM is used to predict altruistic behaviours, which was proposed by Schwartz (Schwartz, 1977). It is assumed that Personal Norms (PN) (i.e., a moral obligation to conduct a behaviour) are shaped by Awareness of Consequences (AC), which refers to whether a particular behaviour is associated with specific consequences, and Ascription of Responsibility (AR), which refers to the perceptions of responsibility to conduct specific behaviour (Stern et al., 1999). The combined approach has been commonly adopted in other similar studies examining travel behaviour mechanisms and transport policies (e.g., Bamberg et al., 2011; Cheng et al., 2022; Liu et al., 2017; Savari et al., 2023). For example, a study used TPB and NAM to investigate the role of government-supported transport policies in the adoption of low-carbon travel modes in Tianjin, China (Liu et al., 2017). Similarly, a study investigating different factors affecting the acceptability of different transport demand management measures using TPB and NAM was conducted in Lahore, Pakistan (Javid et al., 2015). As human behavior is complex, it can be challenging to gain valuable insights from a single theory to fully explain individual behavioral outcomes. This combined approach is expected to enhance the predictive capacity of the model and provide a more comprehensive understanding of social acceptance of women's mobility. The results can offer valuable insights for the development of interventions to improve women's mobility.

1.4. Research objectives, contributions, and paper outlines

This study employs the conceptual framework of behavioural theories of TPB and NAM to evaluate the social acceptance of women's mobility, using Pakistan as a case study. An online comprehensive questionnaire was administered to collect data, which were analysed utilizing the Confirmatory Factor Analysis (CFA) and the multivariate Structural Equation Modelling (SEM) techniques. The results led to the formulation of pertinent transport policy recommendations aimed at fostering inclusive transport services for women in Pakistan, and more broadly for developing countries sharing similar socio-cultural fabric.

This study makes three main contributions. First, it adopts the combined TPB and NAM approach to study the social acceptance of women's mobility considering the role of socio-cultural dynamics. Second, it highlights the role of Transport Policies (TP) in activating behaviour in society. Third, it suggests numerous transport policies for enhancing transport equity. There are potential implications for policymakers in better understanding the challenges of women's mobility and developing plans to provide equitable transport for women.

The rest of the paper is structured as follows. The design of the questionnaire and the approaches to data collection are elaborated in Section 2. Section 3 shows the data analysis and the methodology of SEM. Section 4 encompasses the presentation of results, whereas discussions and policy implications are elaborated in Section 5. Finally, Section 6 concludes the paper and recommends future research directions.

2. Data

2.1. Data collection

This questionnaire survey took place during May and June of 2023. Due to prevalent socio-cultural norms that discourage unnecessary interactions with women, conducting face-to-face interviews with women respondents was challenging. To collect responses from respondents with diverse backgrounds, an alternative survey method was employed. Respondents were approached through various social media platforms, including Facebook, LinkedIn, and Twitter. The link to the questionnaire survey was shared in various Facebook groups, and multiple LinkedIn and Twitter posts were made to reach a wider spectrum of respondents, especially those women with limited mobility who did not usually go out. At the beginning of the survey, a description of the survey objectives was given, and respondents were informed that their responses would be completely anonymous and used for research purposes only. The survey was kept open for about four weeks to collect responses. In view of the fact that SEM analysis requires a sample size that is at least ten times the number of observed variables (statements) (Nicolaou and Masoner, 2013; Wolf et al., 2013). For this study, 30 observed variables were incorporated, so the recommended sample size should not fall short of 300 responses. To meet this minimum requirement, a total of 478 responses were successfully collected for subsequent analysis.

2.2. Questionnaire design

The questionnaire survey was primarily split into two parts. In the first part, the respondents were asked about their socio-demographic and travel-related features considering the socio-cultural fabric in the country, including gender, marital status, age, education, profession, income, number of household family members, number of working women in the household, household car and motorcycle ownership, frequently used transport mode by women family members, and trip frequency of women family members with this frequently used transport mode.

In the questionnaire's second part, statements were constructed regarding social acceptance (SA) of women's mobility in Pakistan. This study adopts a combined approach of the TPB and NAM to explore the social acceptance of women's mobility.

The TPB asserts that an individual's behaviour is influenced by intentions, and intentions are influenced by their attitudes, SN, and PBC. They are significant predictors of model behavioural intentions (Bae and Chang, 2021; Donald et al., 2014). The behavioural intentions also play the role of mediator in explaining the influence of attitudes, SN and PBC on behaviour. PBC is a direct predictor of behaviour. The influence of attitudes is positive on behavioural intentions. In contrast, PBC and SN can have either positive or negative influences, depending on how researchers define these latent variables as received encouragement or discouragement (Donald et al., 2014; Ali et al., 2023). Influenced by the socio-cultural dynamics, it is assumed that society encourages the social acceptance of women's mobility. Hence, the latent variable of SN has a positive influence on the social acceptance of women's mobility. Conversely, the society conservativeness, social and religious beliefs and safety concerns may discourage women's mobility. The respondents have a higher awareness of these social-cultural constraints which lead to a higher value of the PBC variables. It is assumed that PBC and these constraints negatively influence the social acceptance of women's mobility.

The NAM posits that PN acts as a predictor of behaviour, which is determined by AC of behavioural outcomes, and AR for negative or positive outcomes associated with a particular behaviour. Anticipated feelings of guilt and perceived social norms indicate that PN serves as a significant and positive predictor for the intention to use public transport (Bamberg et al., 2007). It defines the moral obligation of individuals to conduct or support a specific behaviour. Research has showcased that AC and AR can exert a direct or indirect influence on PN (De Groot and Steg, 2009). In this study, we assumed that PN, as a predictor of AC and AR, positively affects the social acceptance of women's mobility based on the consciousness of respondents' responsibility and awareness of the consequences.

To enhance the explanatory power of the TPB and NAM model, we introduced a latent variable of supportive Transport Policies (TP) as an integral component of the model. Previous research has highlighted the impact of different policy measures in improving transport behaviour (Hidayati, 2023; Mattauch et al., 2016). For example, a set of socio-psychological variables and the influence of government support was used to predict intentions of using low-carbon travel modes in Tianjin, China (Liu et al., 2017). Similarly, factors influencing the acceptability of different travel demand management strategies were studied in Lahore, Pakistan (Javid et al., 2015). However, the role of TP as a mediator to improve the social acceptance of women's mobility using a combined TPB and NAM approach has not been explored. Thus, this paper bridges this research gap using the proposed innovative modelling approach. It is assumed that PBC and PN of the respondents have a positive influence on the perceived effectiveness of supportive transport policies, and the transport policies enhance the social acceptance of women's mobility.

Considering the local, social, cultural, and infrastructural factors in Pakistan, the following nine hypotheses are formulated and explored. Fig. 1 depicts the conceptual framework incorporating variables from the TPB and NAM:

H1: The attitudes exert a positive influence on social acceptance of women's mobility.

H2: The subjective beliefs contribute positively to the social acceptance of women's mobility.

H3: The awareness of behavioural control variables and social-cultural constraints positively impact the social acceptance of women's mobility.

H4: The awareness of behavioural control variables and social-cultural constraints have a positive influence on the perceived effectiveness of supportive transport policies to enhance women's mobility.

H5: The awareness of consequences positively influences the development of ascription of responsibility among respondents related to their social behaviour and norms.

H6: The sense of responsibility among respondents has a positive influence

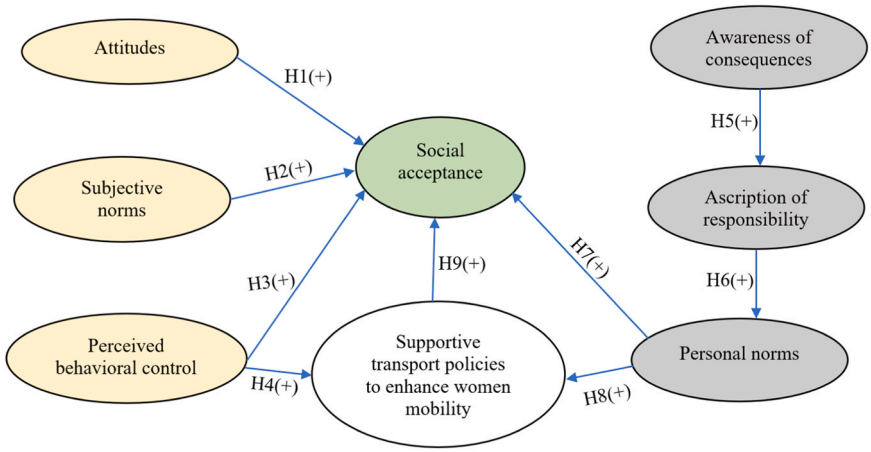

Fig. 1. Conceptual framework with TPB and NAM variables.

on the activation of their norms related to women's mobility.

H7: The personal norms of travellers have a positive influence on social acceptance of women's mobility.

H8: The personal norms of travellers have a positive influence on the perceived effectiveness of supportive transport policies to enhance women's mobility.

H9: The perceived effectiveness of supportive transport policies to enhance women's mobility has a positive influence on social acceptance.

First, about the TPB variables, three statements were designed in connection with the latent variables of attitudes. Sometimes, family beliefs influence the socio-cultural values which directly impact attitudes towards certain behaviours. In this case, these statements are designed considering respondents' family conservativeness, risky road environment, consciousness about women traveling, the importance of family and social values, independent traveling of women, women's driving skills ability, and permission for women to travel alone (Khan, 1999; Adeel et al., 2017; Teng et al., 2023; Zulfiqar, 2020):

I believe in the independent travelling of women (Attitude-1)

I believe that women are equally good at driving (Attitude-2)

In our family, women are allowed to travel alone on public transport (Attitude-3)

Four statements for SN latent variable were designed considering the subjective beliefs of the respondents towards women's mobility. These statements are concerned with the acceptance of independent traveling of women, permission to travel with friends, the influence of social values on women traveling alone, and society's openness towards women's traveling and driving alone (Adeel et al., 2017; Al-Garawi and Kamargianni, 2022; Bamberg et al., 2007):

Women are allowed to travel alone when they are travelling with their friends (SN-1)

It is acceptable in society for women to drive a vehicle (SN-2)

Social values usually influence women's travel behaviour (SN-3)

Society is more open to the independent travelling of women (SN-4)

The PBC variable consisted of three statements that consider society conservativeness, social and religious values, and women's security as the main constraints restricting the women's ability to travel and drive alone (Oktay et al., 2023; Hidayati, 2023; Hosseini et al., 2022):

Conservative society is a hurdle in women alone travelling (PBC-1)

Social/religious values are the main barrier to women's travelling (PBC-2)

Women's security is a main hurdle in their travelling alone (PBC-3)

Second, about the NAM variables, the PN variable consisted of four statements that determine the travellers' moral obligation to support independent women's traveling, provide friendly road environments and endorse their driving (Bamberg et al., 2007; Liu et al., 2017). It is supposed that the sense of awareness and responsibility among travellers regarding their behaviour will help in activating their norms to favour women's traveling and driving alone to support social acceptance of women's mobility:

It is my moral responsibility to support the independent travelling of women (PN-1)

I feel a moral responsibility to endorse women's driving to ensure their proper travelling (PN-2)

I feel a moral responsibility for the social protection of women when they travel on public transport (PN-3)

I feel a moral responsibility to provide a supportive road environment for women drivers (PN-4)

Three statements were designed related to the AC variable which determines the awareness level of the respondents concerning women's mobility freedom, social and religious restrictions, family conservativeness, and road driving environments (Adeel et al., 2017; Oktay et al., 2023):

I know that social and religious constraints restrict women's independent travelling (AC-1)

I know that family conservativeness has a direct link with women's independent travelling (AC-2)

I know that the road driving environment influences women's driving intentions and security (AC-3)

Similarly, three statements were structured for the AR variable regarding road users' sense of responsibility related to the constraints and difficulties faced by the women segment of the travel market (Graglia, 2016). The travellers' awareness about the perceived outcomes of their behaviour which restrict women's mobility can be useful in deriving the transport policies to promote equitable transport:

I feel equal responsibility for an unsecured driving environment for women (AR-1)

I am equally responsible for restricted women's travelling (AR-2)

I am equally responsible for social and cultural constraints related to women's travelling (AR-3)

At last, four statements were constructed to measure the SA variable. The main motives of these statements include support for women's traveling and driving regardless of the social and cultural values predominant in the community (Adeel et al., 2017), favours for independent travelling, acceptance of the women's right to drive, and women's travelling alone by public transport:

I am always supportive of women's driving regardless of social and cultural values (SA-1)

I am the kind of person who always favours the independent travelling of women (SA-2)

I believe that women have the right to travel alone to meet their daily needs (SA-3)

I am supportive of women's travelling alone by public transport modes (SA-4)

There were six statements concerning the evaluation of the TP

variable that can help to enhance women's mobility. The selected transport policies include preferential parking for women riders/drivers, supportive road environments for women drivers through education and awareness of road users, dedicated space on public transport vehicles, support for carpooling/carsharing among women riders, and dedicated public transport buses:

Preferential parking for women drivers/riders (TP-1)

Supportive road environment for women drivers/riders through education and awareness (TP-2)

Dedicated space in a public transport vehicle for women (TP-3)

Support for carpooling/carsharing among women riders (TP-4)

Support from traffic enforcement agencies for women (TP-5)

Separate public transport buses for women (TP-6)

All the statements regarding the TPB and NAM variables were assessed utilizing a five-point Likert scale to gauge the level of agreement, ranging from strongly disagree (1), disagree (2), neutral (3), agree (4), to strongly agree (5). The use of a five-point Likert scale provided a better understanding of the collected responses and could yield a satisfactory level of reliability among responses (Croasmun and Ostrom, 2011).

3. Methodology

3.1. Validity and reliability tests

The collected data underwent a thorough examination and validation, utilizing various statistical parameters. Initially, the Kaiser-Meyer-Olkin (KMO) and Bartlett's tests were performed to assess the adequacy of the collected samples. A KMO value exceeding 0.7 and Bartlett's test significance level below 0.05 suggests that the data is suitable for factor analysis (Taber, 2018). Subsequently, Cronbach's alpha, Composite Reliability (CR), and Average Variance Extracted (AVE) were calculated. A Cronbach's alpha exceeding 0.7, a CR value surpassing 0.7, and an AVE value exceeding 0.5 indicate satisfactory levels of reliability and convergent validity of the collected data (Fornell and Larcker, 1981). Finally, the Skewness and Kurtosis values were computed to assess the normality of the collected data. If their associated values fall within a range from −3 to +3, and a range from −10 to +10, respectively, the data is confirmed to be normally distributed, and suitable for use in SEM analysis. The factor loadings were calculated to verify the reliability of the data and affirm the correlation between observed variables and their respective latent variables or factors, as outlined in the hypothesis during the questionnaire design stage, as illustrated in Fig. 1. A factor loading surpassing 0.7 suggests a strong association of observed variables with unobserved variables (Taber, 2018). An observed variable with higher factor loadings is presumed to carry more significance in explaining the associated unobserved or latent variable. Taking into consideration that people's reactions to women's mobility might vary significantly based on gender, an ANOVA analysis was also conducted to determine any significant differences among respondents in their perceptions of women's mobility.

3.2. Structural equation modelling

SEM is a well-established method used for the analysis of multivariate data. It can handle a considerable number of both endogenous and exogenous variables, including latent (unobserved) variables, which are defined as linear combinations of observed variables. It is a powerful analysis technique that can be used to conduct multivariate analysis between different observed and unobserved variables. It has been widely adopted in numerous researches to examine the travel behaviour of commuters for exploring theory-driven structural hypotheses (e.g., Ali et al., 2023; de Oña et al., 2013; Li et al., 2023; Liu et al., 2022). It consists of two components: a measurement model and a structural model. The measurement model is employed to assess the unobserved latent variable, which cannot be directly measured and is typically a linear function of the observed variables. On the other hand, the structural model depicts the direction and strength of relationships among various latent variables. In a generic form, the measurement model can be represented as follows (Bollen, 1989):

$$x = \Lambda_x \xi + \delta \tag{1}$$

and

$$y = \Lambda_y \eta + \varepsilon \tag{2}$$

where x and y are the column vectors of the observed exogenous variable and endogenous variable, respectively. Λ_x and Λ_y are the structural coefficient matrix of the unobserved latent exogenous variable (ξ) and endogenous variable (η) on the observed variables, respectively. δ and ε are the residual vectors of the exogenous variable and endogenous variables, respectively.

The generic formula for representing the structural model is as follows:

$$\eta = B\eta + \Gamma\xi + \zeta \tag{3}$$

where B is a random connection matrix reflecting the relationship between endogenous variables, Γ is a direct random effect matrix, illustrating the impact of exogenous variables on endogenous variables and ζ is the residual vector of the structural model.

Because of the prevalent use of SEM, different estimation methods are used for developing structural equation models. The maximum likelihood method stands out as the most popular approach, often integrating robust chi-squares and standard errors to account for the unmodeled heterogeneity when estimating the structural equation system (Golob, 2003), which was applied in this study. In addition, the reliability of the developed structural equation model was assessed by comparing values of the chi-square ratio to the degree of freedom (χ^2/DF), the comparative fit index (CFI), the goodness of fit index (GFI), the root mean square residual (RMR), and the root mean square error approximation (RMSEA) with permissible values. The recommended values are greater than 0.9 for CFI, and GFI, less than 0.08 for RMSEA, and between 2 and 5 for χ^2/DF (Bentler and Bonett, 1980; McKelvey and Zavoina, 1975; Sahoo, 2019).

4. Results

4.1. Descriptive statistics

Table 1 presents descriptive statistics on the socio-demographics and travel-related features of the respondents. The distribution shows that almost 37.0 % of the respondents were women which represents a good share of the sample considering their share in the working population. This travel share is consistent with the findings of previous studies (Adeel et al., 2017; Khan, 1999). Around 53.8 % of the respondents were in the age group of 21–30 years, whereas under 20 years and above 30 years segments had a share of around 23.0 % and 23.2 %, respectively. The sample distribution shows that the collected samples mainly comprised young travellers. These statistics are consistent with the results of a recent consensus conducted in the country (Pakistan Bureau of Statistics, 2017). Most of the respondents were either graduate or undergraduate students. The majority of them had a family monthly income of less than 50,000 Pakistani Rupees (about 280.2 Pakistani Rupees are equivalent to 1 US dollar). Around 55.6 % of the respondents had a family size of 4–6 members, and there were no working women members in half of the households. The household car ownership and motorcycle ownership shares were almost 54.4 % and 83.9 %, respectively. The women's modal split shows that the majority of the women depended on private transport, where most were picked-up/dropped-off by their male family members, and only a quarter used public transport. It implies that women's mobility was quite restricted and highly

Table 1
Descriptive statistics of respondents' socio-demographics and travel-related features.

Socio-demographic characteristics and travel-related features	Categories	Frequency	Percentage (%)
Gender	Men	301	63.0
	Women	177	37.0
Marital status	Single	341	71.3
	Married	137	28.7
Age (years)	Under 20	110	23.0
	21–30	257	53.8
	Above 30	111	23.2
Education	Bachelor and below	333	69.9
	Master and above	145	30.1
Profession	Student	274	57.3
	Employee	153	32.1
	Business	16	3.3
	Others	35	7.3
Family monthly income (Pakistani Rupee)	Under 50,000	272	56.9
	50,000–100,000	89	18.6
	Above 100,000	117	24.5
Number of household family members	1–3	76	15.9
	4–6	266	55.6
	More than 6	136	28.5
Number of working women family members in the household	0	234	49.0
	1	140	29.3
	2 or above	104	21.7
Household car ownership	0	218	45.6
	1	200	41.8
	2 or above	60	12.5
Household motorcycle ownership	0	77	16.1
	1	252	52.7
	2 or above	149	31.2
Frequently used transport mode of women's family members	Drive a motorcycle	22	4.6
	Drive a car	56	11.7
	Pick up/drop off	242	50.2
	by others	108	22.6
	Take public transport	52	10.9
	Others		
Trip frequency of women family members with the above mode (in a week)	Less than once	242	50.6
	1–2 days	70	14.6
	3–4 days	36	7.5
	5–6 days	58	12.1
	Every day	71	14.9

dependent on others. Previous research also reported a lower share of women in travel as compared to men (Abdullah et al., 2021; Javid et al., 2016). The results indicate that a significant portion of the women had a very low trip frequency, as 50.6 % of them travelled less than once a week, and only 51.0 % of the households had at least one working woman.

4.2. Results of validity and reliability tests

Table 2 displays the results of the KMO and Bartlett's test of sphericity. It is noticed that the estimated KMO value significantly exceeds 0.7, and the level of significance for Bartlett's test is well below 0.05, both of these results satisfy the criteria (Shrestha, 2021). The findings confirm that the degree of information among latent variables overlaps marvellously, which ensures that an excellent correlation exists between latent variables. Therefore, the data is suitable for conducting factor analysis. The p–value being less than the significant level of 0.05 confirms the rejection of the null hypothesis, indicating that the correlation matrix is not an identity matrix.

Table 2
KMO and Bartlett's test of sphericity results.

KMO and Bartlett's test		Results
KMO measure of sampling adequacy		0.903
Bartlett's test of sphericity	Approx. Chi-square	9098.5
	Degree of freedom	561
	Level of significance	<0.001

The values of Cronbach's alpha, CR, and AVE of the latent variables are tabulated in Table 3. Comparatively, the latent variables of PN, SA, and TP have more influence in explaining the latent variables with Cronbach's alpha values of 0.891, 0.898, and 0.910, respectively. The analysis revealed that all the latent variables, except SN and PBC, have high reliability with CR values higher than 0.7 (Fornell and Larcker, 1981). Additionally, PN, AC, AR, SA, and TP show a high convergence as their AVE values are higher than 0.5 (Hair, 2009). Although attitude, SN, and PBC do not meet both requirements, the convergence validity of these constructs is considered acceptable given their CR values exceeding 0.6 and AVE values exceeding 0.4 (Fornell and Larcker, 1981; Li et al., 2018).

The findings of average responses, reliability and normality tests, factor loadings, and one-way ANOVA are summarized in Table 4. The average responses for most of the observed variables of attitudes and SN latent variables are above 3.0 (representing from neutral to agree on the five-point Likert scale). Most of the respondents agreed that they believe in women's freedom in travelling and it is acceptable for them and in society when women travel alone and drive a vehicle. The mean score of the PBC variable is 3.441 which depicts a higher level of respondents' awareness about the perceived social and cultural constraints in women's mobility. The latent variables of PN, AC, and TP have higher agreement compared to the other latent variables with a mean score of 3.84, 3.66 and 3.79, respectively. These mean scores predict a higher level of awareness and moral obligations among respondents about the constraints of women's travelling in society. Also, respondents showed a higher agreement for the support of relevant transport policies to promote women's mobility. The mean score of the SA variable is 3.56 which shows high agreement among respondents and social acceptance of women's independent travelling. The latent variable of AR showed the lowest agreement of the respondents with a mean score of 3.07, which depicts a low sense of responsibility among some respondents regarding social-cultural constraints regarding women's mobility. The normality tests were conducted on responses, and Skewness and Kurtosis values were estimated. The Skewness values range from −1.320 to +0.137 and are within the acceptable range from −3 to +3 for SEM analysis. Besides, the Kurtosis values range from −1.129 to 1.480, which lie within a range from −10 to +10 for SEM analysis (Brown, 2006). The data is normally distributed for use in SEM analysis. Employing the maximum likelihood method, the factor loadings were calculated to affirm the relationship between observed variables and their respective latent variables as defined in the questionnaire design stage based on the TPB and NAM theories. A factor loading threshold value of 0.5 was employed as a cut-off point for extracting the factor loading. All the defined variables confirmed their association with latent variables with a factor loading above 0.5. Variables with higher factor loading exert a stronger explanatory influence on the corresponding latent variable. An ANOVA analysis was conducted to compare among different categories of the respondents based on their gender. The ANOVA results for gender depict that there is a significant difference between men and women respondents regarding perceptions about women's mobility except for

Table 3
The values of Cronbach's alpha, CR, and AVE of the latent variables.

Latent variable	Cronbach's alpha	CR	AVE
Attitude	0.726	0.731	0.478
SN	0.688	0.695	0.417
PBC	0.669	0.672	0.408
PN	0.891	0.886	0.665
AC	0.843	0.846	0.648
AR	0.807	0.828	0.629
SA	0.898	0.901	0.694
TP	0.910	0.911	0.634

Table 4
Average responses and standard error, Skewness and Kurtosis values, factor loadings, and one-way ANOVA results.

Observed variables	Mean	Standard error	Skewness	Kurtosis	Factor loadings	One-way ANOVA *Gender* (*F-value*)
I believe in the independent travelling of women (Attitude−1).	3.462	0.057	−0.452	−0.836	0.764	42.885**
I believe that women are equally good at driving (Attitude−2).	3.236	0.057	−0.270	−0.892	0.715	61.000**
In our family, women are allowed to travel alone on public transport (Attitude−3).	3.104	0.059	−0.201	−1.129	0.584	9.825**
Women are allowed to travel alone when they are travelling with their friends (SN−1).	3.071	0.058	−0.298	−0.986	0.615	15.962**
It is acceptable in society when women to drive a vehicle (SN−2).	3.477	0.052	−0.629	−0.241	0.703	1.712
Social values usually influence women's travel behaviour (SN−3).	3.500	0.048	−0.748	0.110	0.580	4.532*
Society is more open to the independent travelling of women (SN−4).	2.807	0.054	0.137	−0.881	0.504	0.534
Conservative society is a hurdle in women alone travelling (PBC−1).	3.506	0.057	−0.530	−0.693	0.715	13.370**
Social/religious values are the main barrier to women's travelling (PBC−2).	3.058	0.058	−0.136	−1.000	0.568	0.526
Women's security is a main hurdle in their travelling alone (PBC−3).	3.759	0.055	−0.820	−0.180	0.625	2.423
It is my moral responsibility to support the independent travelling of women (PN−1).	3.592	0.053	−0.722	−0.166	0.664	23.757**
I feel a moral responsibility to endorse women's driving to ensure their proper travelling (PN−2).	3.719	0.050	−0.788	0.067	0.747	12.049**
I feel a moral responsibility for the social protection of women when they travel on public transport (PN−3).	4.041	0.048	−1.320	1.483	0.925	4.913*
I feel a moral responsibility to provide a supportive road environment for women drivers (PN−4).	4.041	0.048	−1.269	1.326	0.897	2.818
I know that social and religious constraints restrict women's independent travelling (AC−1).	3.579	0.051	−0.593	−0.254	0.789	2.428
I know that family conservativeness has a direct link with women's independent travelling (AC−2).	3.665	0.050	−0.738	0.056	0.883	7.024**
I know that the road driving environment influences women's driving intentions and security (AC−3).	3.757	0.046	−0.806	0.378	0.736	0.869
I feel equal responsibility for an unsecured driving environment for women (AR−1).	3.297	0.053	−0.328	−0.667	0.523	3.092
I am equally responsible for restricted women's travelling (AR−2).	2.968	0.054	−0.097	−0.904	0.852	0.467
I am equally responsible for social and cultural constraints related to women's travelling (AR−3).	2.997	0.055	−0.124	−0.905	0.943	0.977
I am always supportive of women's driving regardless of social and cultural values (SA−1).	3.543	0.053	−0.692	−0.264	0.752	30.251**
I am the kind of person who always favours the independent travelling of women (SA−2).	3.527	0.055	−0.604	−0.468	0.870	44.362**
I believe that women have the right to travel alone to meet their daily needs (SA−3).	3.713	0.054	−0.845	−0.070	0.892	32.468**
I am supportive of women travelling alone by public transport modes (SA−4).	3.491	0.056	−0.579	−0.541	0.812	39.038**
Preferential parking for women drivers/riders (TP−1).	3.520	0.054	−0.684	−0.360	0.729	10.896**
Supportive road environment for women drivers/riders through education and awareness (TP−2).	3.895	0.049	−1.130	0.807	0.844	7.770**
Dedicated space in a public transport vehicle for women (TP−3).	3.866	0.054	−1.031	0.268	0.845	6.498*
Support for carpooling/carsharing among women riders (TP−4).	3.690	0.051	−0.743	−0.053	0.804	6.080*
Support from traffic enforcement agencies for women (TP−5).	3.853	0.050	−1.073	0.716	0.834	3.339
Separate public transport buses for women (TP−6).	3.907	0.052	−0.959	0.164	0.711	2.835

Note: **significant at the 1 % level (*p-value* < 0.01),
*significant at the 5 % level (*p-value* < 0.05)

variables where the results are insignificant (Table 4). The difference in response means of men and women shows that their perceptions differ significantly for most of the observed variables. All the constructs of attitude and SA are significant at the 1 % level. The findings concur with another similar study which shows that gender manifests different behaviour regarding women's mobility (Lodhi et al., 2022). While the responses of men and women regarding constructs of AR are not statistically significant at the 1 % or 5 % level.

4.3. Results of structural model

The structural model, constructed through SPSS AMOS software and the maximum likelihood method, depicts the outcome of social acceptance incorporating variables from the TPB and NAM, as illustrated in Fig. 2. Only significant structural relationships are shown in Fig. 2. The structural model shows the influence of the TPB and NAM variables and the effect of transport policy interventions toward social acceptance of women's mobility.

The attitudes exhibit a strongly positive and significant structural relationship with SA ($\beta = 0.88$, $p < 0.01$), which shows that the people with positive beliefs concerning women's independent travelling, travelling alone on public transport and their driving skills have more social acceptance related to their mobility. The association between SN and SA is negatively significant at the 1 % level ($\beta = -0.38$, $p < 0.01$). This suggests the presence of respondents whose beliefs are influenced by social pressure despite they have acceptance of women's mobility choices. It can be argued that sometimes the specific responses are subjected to social pressure. The structural relationship of PBC with SA was insignificant. However, the structural estimate of the PBC variable with the TP variable is positive and significant ($\beta = 0.17$, $p < 0.01$). This suggests that, despite firm beliefs on social acceptance and other constraints in women's travelling, some people perceived that the supportive transport policies have the potential to safeguard women's travelling as indicated by the positive structural coefficient between PBC and TP.

The variables of AC have a positive and significant relationship with those of AR ($\beta = 0.27$, $p < 0.01$). It predicts that there are travellers who have an awareness of the problems facing women when traveling and take responsibility for the hurdles and constraints in their mobility due to various socio-cultural boundaries. The sense of responsibility positively influences the cultivation of moral obligations among individuals regarding women's freedom and independence in their travel choices as the structural coefficient of AR with PN is positive and significant ($\beta = 0.23$, $p < 0.01$). The variable of PN has positive and highly significant relationships with SA ($\beta = 0.31$, $p < 0.01$) and TP ($\beta = 0.61$, $p < 0.01$) which depict that the people's moral obligation to protect women in

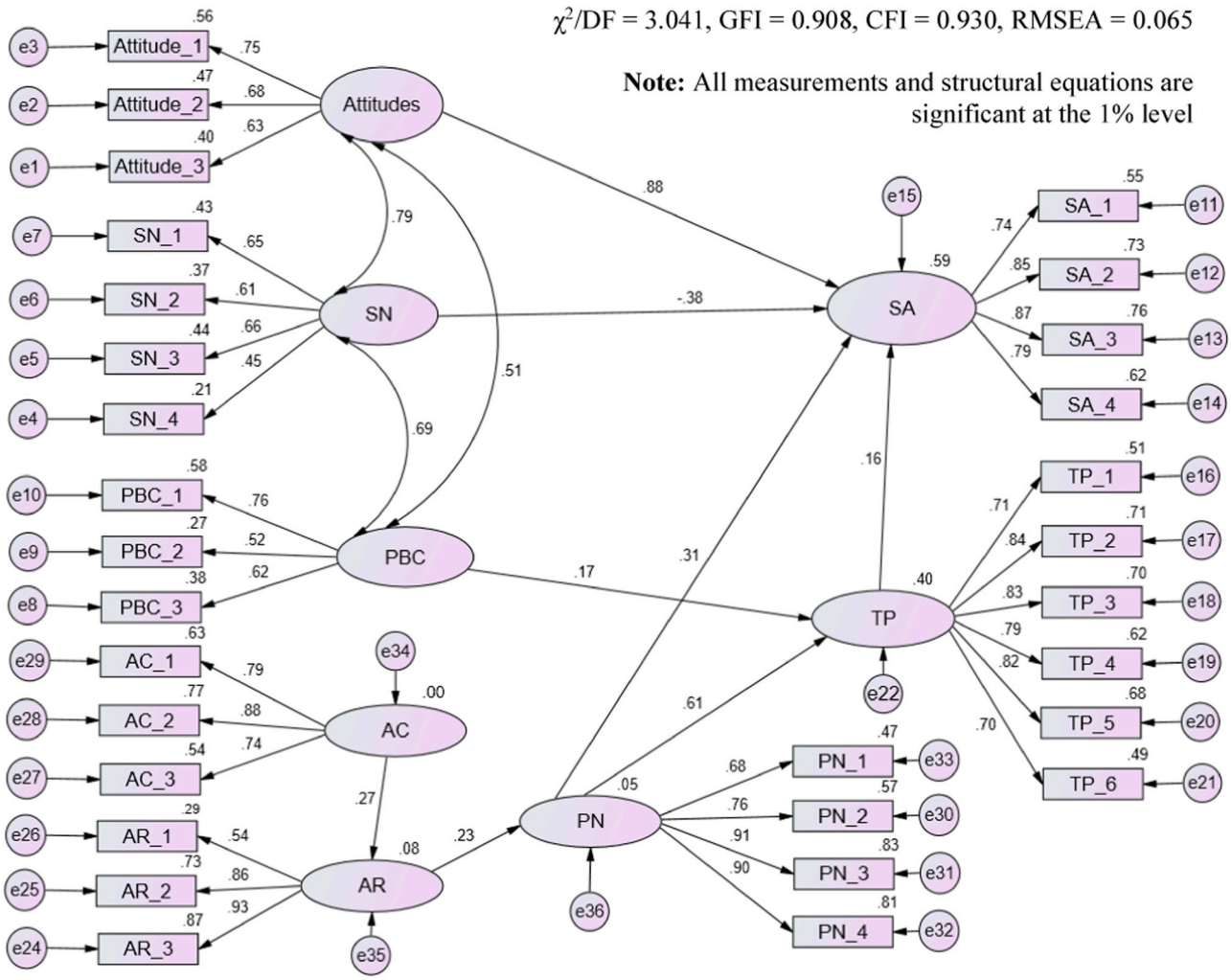

Fig. 2. Structural model of social acceptance of women's mobility with TPB and NAM variables.

driving and providing them independence in travelling is helpful to make women's mobility socially acceptable, and integration of relevant transport policies can boost their social acceptability. The attitudes, SN, PN, and TP variables explain almost 59 % of the variability in SA, whereas the PBC and PN explain almost 40 % of the variance in TP. The structural estimate between TP and SA is significant at the 1 % level and positive ($\beta = 0.16$, $p < 0.01$). This suggests that the stated transport policies such as preferential parking for women riders, dedicated bus service, and carpooling programs, public awareness and support about gender equality, and support from traffic enforcement agencies can help to step up the social acceptance of women's mobility. The perceived transport policies have a positive impact on improving the socio-cultural acceptance of women's mobility because of the positive and significant structural relationship between TP to SA.

The values of the goodness of fit parameters fall within the ranges of acceptance. For instance, the chi-square ratio to the degree of freedom (χ^2/DF) is 3.041, which is between 2 and 5, GFI and CFI are 0.908 and 0.930 respectively, both are higher than 0.9. The RMSEA is 0.065, which is less than 0.08 (Hooper et al., 2008). The results indicate that the constructed structural model has good reliability in explaining the social acceptance of women's mobility.

It is believed that the individual responses based on gender might also manifest significantly different behaviour toward social acceptance of women's mobility. Therefore, in addition to the base model, two separate models for men and women were developed to determine their perceptions about women's mobility. The results of the hypothesis testing are tabulated in Table 5. It is noted that the hypotheses H1, H4, H5, H6, H7, H8, and H9 are not rejected because they are significant, and signs of structural estimates were consistent with the expected signs. The attitudes toward women's mobility have a positive impact on social acceptance of their mobility choices (H1) and the value of the path coefficient is 0.88. It infers that the attitudes of the respondents towards

Table 5
Hypotheses related to social acceptance of women's mobility.

Hypothesis	Structural equations			Base model	Men model	Women model
H1	SA	←	Attitudes	Do not reject	Do not reject	Do not reject
H2	SA	←	SN	Reject *(opposite sign)*	Reject *(opposite sign)*	Reject
H3	SA	←	PBC	Reject	Reject	Reject
H4	TP	←	PBC	Do not reject	Do not reject	Reject
H5	AR	←	AC	Do not reject	Do not reject	Do not reject
H6	PN	←	AR	Do not reject	Do not reject	Do not reject
H7	SA	←	PN	Do not reject	Do not reject	Do not reject
H8	TP	←	PN	Do not reject	Do not reject	Do not reject
H9	SA	←	TP	Do not reject	Do not reject	Do not reject

women's mobility positively affect the social acceptance of women to travel and drive independently. The PBC variable has a positive and significant structural estimate with the TP variable (H4) with a path coefficient of 0.17. It depicts that the respondent's positive or strong beliefs on perceived barriers and constraints about women's mobility positively influence their perceptions about the supporting transport policies. The AC has a positive influence with a path coefficient of 0.27 on developing ascription of responsibility among the respondents related to their social behaviour and norms (H5). Hence, it can be concluded that the respondents are cognizant of the consequences of their actions and believe they have the responsibility to enhance social acceptance regarding women's mobility. The sense of responsibility among respondents has a positive impact with a path coefficient of 0.23 on the activation of their norms related to women's mobility (H6), referring that sense of responsibility can positively impact their social norms about women's mobility. The personal norms of travellers' have a positive impact on social acceptance of women's travelling (H7) with a path coefficient value of 0.31. It can be said that the respondents have personal consciousness towards social acceptance of women's mobility in the given circumstances. The positive impact of personal norms of travellers on the perceived effectiveness of supportive transport policies to enhance women's mobility (H8) is evident, as indicated by a path coefficient value of 0.61. This means that most of the respondents hold the belief that supportive transport policies have the potential to enhance social acceptance of women's mobility. Therefore, it can be reasoned that respondents have a strong belief that supportive transport policies can effectively improve social acceptance of women's mobility. The perceived effectiveness of supportive transport policies to enhance women's mobility has a positive impact with a path coefficient value of 0.16 on social acceptance of mobility choice (H9). The confirmation of this hypothesis suggests that supportive transport policies can significantly enhance the social acceptance of women in the community. Considering the positive impacts of personal norms, it can be confirmed that supportive transport policies can play a pivotal role in enhancing the social acceptance of women's mobility.

The hypotheses H2 and H3 are rejected because SN has a negative relationship with SA (H2) and the structural relationships of PBC and SA (H3) are insignificant in all three models. It infers that the role of social values is negative and PBC does not directly affect the social acceptance of women's mobility. The hypothesis H2 for the base and men model is rejected despite significant structural equations because the signs of the structural equations are opposite to the expected sign. It infers that men respondents have significantly different normative behaviour towards social acceptance of women's mobility and supportive transport policies as compared to their counterparts. Hypothesis H4 for the women model is also rejected because the structural equation is insignificant. Therefore, it can be revealed that PBC does not significantly affect the social acceptance of women's mobility.

5. Discussions and policy implications

Within the frameworks of the TPB and NAM, this study innovatively adds the role of TP to discuss the psychological mechanisms of respondents regarding the social acceptance of women's mobility. It has been discovered that attitudes exert a positive and significant influence on social acceptance. This aligns with previous research, which has consistently demonstrated the substantial impact of attitudes on behaviour (Chen and Chao, 2011; Chen et al., 2019). This suggests that positive attitudes within the community, such as supporting women traveling alone on public transport and believing in their driving skills, can enhance the social acceptance of women's mobility. These attitudes can create a satisfactory and comfortable environment for women on public transport and foster a supporting culture on roads. In contrast, the relationship between SN and SA is negative and significant which demonstrates that the conservativeness in society and social restrictions will continue to limit women's mobility unless properly documented policy interventions are not proposed to improve their travel share on public transport and independent driving (Macias-Alonso et al., 2023; Porrazzo et al., 2022). The negative perceptions of the community and restrictive beliefs about women's independent mobility will continue to be a major barrier unless progressive policies are not implemented in true spirit to challenge societal norms.

The results of the structural model reveal a significant positive relationship between AC and AR. This suggests that those who acknowledge the challenges associated with women's mobility are more likely to take responsibility for these problems due to social and cultural barriers. This result validates that survey respondents are aware of the issue and sense a moral and ethical duty to improve the social acceptance of women's mobility by addressing the behaviour that leads to social restrictions. It is found that AR is positively associated with PN, highlighting the importance of raising awareness about the challenges faced by women in mobility and the need for individuals to take responsibility for addressing these challenges. Developing a sense of responsibility among road users is crucial in building moral obligations and promoting social acceptance of women's mobility, including independent travel, driving, and ridesharing. These findings align with previous research, which has demonstrated a positive correlation between AR and PN (Javid et al., 2015; Liu et al., 2017). These findings underscore the significance of creating awareness about the discrimination faced by women during travel and the need for individuals to feel responsible for addressing such problems. PN can modify individual's self-conscious perceptions, and prompt them to alter their behaviour. Therefore, the outcomes of this study suggest that AC and AR activate respondents' PN on moral obligations concerning the freedom and independence of women in making their mobility decisions. In other words, respondents with feelings of moral obligation can play a pivotal role in creating a sense of awareness among individuals concerning women's mobility needs.

PN reflects the positive willingness of the respondents towards TP and social acceptance of women's mobility. Addressing these issues requires implementing countermeasures that recognize the complexities of women's needs for education, healthcare, and employment as well as their fundamental right to mobility and independent access to various destinations (Akyelken, 2020). By doing this, not only will their travel share improve, but their contribution to sustainable and liveable societies, where women can play an active role, will also be enhanced. The outcomes of the developed SEM model are consistent with the NAM model, where PN exhibits a significant and positive role in activating pro-social behaviour in the society. The concept of PN refers to individuals' internalized values and beliefs which can guide their behaviour. Activating PN can be a powerful tool as it encourages individuals to recognize the necessity of equal access to mobility needs of women in society. Those with higher PN perceive TP as an effective tool in enhancing women's mobility, which is positively associated with social acceptance. This positive perception is important in demonstrating the tangible benefits of TP in creating equitable transport for women. Therefore, TP should be considered during the planning and policy-making stage, such as awareness campaigns, improving the safety and security of women through dedicated public transport services and active participation of women in planning, and a dedicated carpooling system with preferential parking.

The results showed that people's awareness of perceived barriers significantly influences their support for transport policies required to promote women's mobility. It means that the people who are more aware of the constraints would have high support for relevant transport policies. Awareness campaigns targeting both travellers and transport agencies have the potential to address the harassment and violence barriers that restrict women's mobility on public transport vehicles (Ceccato, 2017; Desai et al., 2021; Kaygan et al., 2022; Tripathi et al., 2017). There is widespread recognition that increasing society's awareness and sense of responsibility for addressing the challenges of women and finding effective solutions can improve their mobility.

Achieving this requires educating the public about women's rights to access various places and eliminating discrimination in the use of public transport and independent driving. This will also help to alter the attitudes and reduce the perceived barriers that hinder women's mobility and independent travelling.

Improving personal security and road safety for women is crucial to increasing their share of public transport use, as cases of harassment can undermine efforts to encourage women to travel (Gardner et al., 2017). One recommendation is to implement measures that enhance women's safety on roads. Providing public transport services that cater specifically to women's traveling needs can help address the issues of gender-based discrimination and harassment in travel (Graglia, 2013; 2016). Implementing gender-responsive transport policies and active participation of women in the transport sector can help address the constraints in the transit environment and culture related to women's mobility and harassment (Mubarok and Suparman, 2019; Noor and Iamtrakul, 2023). Women are more aware of the specific needs related to women's mobility, and their unique experiences and heightened understanding of the specific requirements can contribute to the development of effective policy interventions. By promoting more inclusive and equitable transport for women on the move, they can play an instrumental role in enhancing the social acceptance of women's mobility.

A dedicated carpooling program is recommended to integrate with preferential parking for women, which can encourage mobility sharing and promote sustainable travel behaviour. Carpooling is gaining attention among commuters and provides high aspirations for the users (Prieto et al., 2017). It allows people to use a car without purchasing one, resulting in substantial savings. Moreover, carpooling is more cost-effective compared to personalized travel in a private car (Javid and Ali, 2020). It is worth mentioning that the potential benefits of carpooling are even greater when seat occupancy is high (Ostrovsky and Schwarz, 2019). There are multiple reasons that women may be attracted to the idea of carpooling because of the unique challenges in women's mobility. For example, women's vehicle ownership, education, and awareness disparities are often lower than men. Private vehicle ownership among women is lower due to combined cultural norms and financial constraints in developing countries. Most of the women are not economically independent, limiting their travel to personalized cars, which makes carpooling an attractive option for shared mobility. If society encourages and supports women's driving and independent travel, raises awareness and education about women's mobility, and adequately addresses perceived concerns of safety and harassment, the use of carpool among women can be significantly enhanced.

On the other hand, implementing preferred parking policies for women with shared mobility can significantly improve their travel share. A study found that different incentivized parking policies can modify the commuter behaviour from 10 % to 30 % (Litman, 2016). Additionally, one carpool can reduce the need for at least one other parking spot (Olsson et al., 2019). Providing dedicated parking facilities is an effective way to enhance carpooling and encourage women to share rides. However, it is crucial to emphasize that without proper management and enforcement, the positive impact of preferential parking for carpooling among women may diminish. It is essential to ensure that preferential parking is used as an incentive and is properly utilized by women carpool groups with shared mobility only. This can alleviate the financial burden of parking costs (Shaheen et al., 2016) and make shared mobility an attractive option for women.

It is important to note that the activation of positive attitudes, subjective beliefs, and awareness about perceived barriers have a strong impact on enhancing women's mobility in Pakistan and other countries with similar socio-economic dynamics. Furthermore, people's awareness and sense of responsibility regarding their behavioural outcomes in society strongly influence on their social support for women's mobility needs. Overcoming these barriers and constraints requires societal and cultural changes, supportive transport infrastructure and services, as well as awareness and enforcement policies. The social acceptance of women's independent travel can only be achieved through changes in people's attitudes, norms and beliefs. The intervention through education and awareness campaigns playing a crucial role in this regard. Additionally, implementing women-specific transport infrastructure and services, such as dedicated parking and ride-sharing, and women-only public transport services with dedicated waiting areas, can prevent violence and harassment against women in public places.

6. Conclusion

In developing countries, socio-cultural barriers often hinder women's mobility. Women's domestic responsibilities, coupled with concerns of personal safety, significantly influence their mobility and ability to travel independently. This study evaluates the social acceptance of women's mobility using the results of a questionnaire survey conducted in Pakistan. The questionnaire items were designed in alignment with the theoretical background of the TPB and NAM theories. Analysis of the survey results indicated variations in perceptions of certain variables based on gender. The collected data were normally distributed and have a satisfactory level of reliability and validity for use in SEM analysis. Average responses showed that there is high agreement among respondents about their attitudes, normative beliefs, and awareness about the perceived barriers to women's mobility in society. Respondents also showed high moral obligations, a sense of awareness and social acceptance for women's mobility choices, and support for relevant transport policies. However, a low agreement was shown among some respondents towards the sense of responsibility about the social and cultural constraints of women's mobility.

The latent variable of attitudes shows a positive significant structural relationship with the social acceptance of women's mobility. It infers that respondents with positive beliefs about independent traveling of women, and traveling on public transport are more socially accepting of women's mobility. However, the relationship between SN and SA is negative, which shows that respondents' responses about the social acceptance of women's mobility are subject to social pressure. The relationship between PBC and SA is insignificant. While PBC forms a positive and significant relationship with the TP variable which confirms that despite constraints in women's traveling, respondents perceive that supportive transport policies can enhance women's mobility. The variables of PN and TP also positively influence the social acceptance of women's mobility. These relationships suggest that respondents' moral obligation to protect women's independent traveling is helpful in improving social acceptance of women's mobility and launching associated supportive transport policies.

Numerous transport policies are recommended, such as dedicated bus service or facilities for women, active participation of women for gender-responsive transit solutions, dedicated carpooling programs, and preferential parking for women riders can help promote women's mobility and access to various destinations. Road users' awareness and sense of responsibility which is activated through awareness campaigns and enforcement from concerned authorities can help to improve the security of women travellers and reduce the risks of harassment and violence.

In addition to the TPB and NAM, other theoretical approaches are also possible to apply to this research area, such as using the Value Belief Norm theory (Stern, 2000) to evaluate the pro-environmental behavior of women's mobility choices. Therefore, future research should consider using alternative theoretical approaches to gain a more comprehensive and in-depth understanding of the social acceptance of women's mobility from diverse perspectives.

CRediT authorship contribution statement

Muhammad Ashraf Javid: Writing – original draft, Investigation, Conceptualization. **Nazam Ali:** Writing – original draft, Software,

Methodology, Investigation, Data curation, Conceptualization. **Muhammad Abdullah:** Writing – original draft, Investigation, Conceptualization. **Ryan Cheuk Pong Wong:** Writing – review & editing, Visualization, Supervision.

Declaration of Competing Interest

The authors have no conflict of interests to declare

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